

# Evidence-Based Guidelines for Writing Discussion Sections in Geoscientific Research Papers

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## Introduction

*Discussion sections in geoscientific research papers interpret results within broader scientific contexts and assess study limitations, but many researchers lack clear guidance on how to write them effectively. Evidence-based practices can help improve the quality and impact of these critical sections. (LLM Memory)*

The discussion section serves as the interpretive heart of geoscientific research papers, where authors move beyond presenting results to explain their meaning and significance (Model-Generated). Unlike other sections that follow more standardized formats, discussion sections often pose challenges for researchers because they require synthesizing complex findings with existing literature while addressing uncertainties and limitations (Model-Generated). In geosciences, this challenge is particularly acute given the field's interdisciplinary nature, where studies may span geology, chemistry, physics, and environmental science (Model-Generated). Many early-career researchers and even experienced scientists struggle with structuring discussions effectively, leading to sections that either repeat results without interpretation or make claims that exceed what the data can support (Model-Generated). Evidence-based guidelines can provide a framework for writing discussions that clearly communicate scientific insights, acknowledge uncertainties appropriately, and place findings within the broader context of geoscientific knowledge (Model-Generated).

## Structure and Organization of Discussion Sections

*Discussion sections should connect all research questions and results together in a coherent narrative, typically without subheadings, and focus on interpretation rather than restating conclusions. (4 sources)*

The primary organizational principle for discussion sections centers on integrating all research questions and results into a unified narrative that addresses how findings relate to the original research objectives ([Zelst et al., 2021](#)) ([Zelst et al., 2022](#)). This approach ensures that discussions move beyond isolated observations to create meaningful connections between different aspects of the study. Unlike methods and results sections, which benefit from topical subheadings to organize complex information, discussion sections should typically remain succinct and avoid subheadings to maintain narrative flow ([Jara et al., 2021](#)). This structural choice encourages authors to weave their interpretations together rather than compartmentalizing different aspects of their findings. A clear distinction should be maintained between the discussion and conclusions sections, with the bulk of interpretive content belonging in the discussion rather than being relegated to conclusions ([Mchedlishvili et al., 2022](#)). This organizational framework helps ensure that discussions fulfill their intended purpose of providing comprehensive interpretation and context for research findings.

## Evidence

(Zelst et al., 2021)

[101 Geodynamic modelling: How to design, carry out, and interpret numerical studies](#)

*"The discussion section relates all the questions in the manuscript together: how do these results relate to the original ques-"*

(Zelst et al., 2022)

[101 geodynamic modelling: how to design, interpret, and communicate numerical studies of the solid Earth](#)

*"The discussion section relates all the questions in the manuscript together: how do these results relate to the original ques-"*

(Jara et al., 2021)

[Episodic construction of the early Andean Cordillera unravelled by zircon petrochronology](#)

*"The Results and Methods sections should be divided by topical subheadings; the Discussion should be succinct and may not contain subheadings."*

(Mchedlishvili et al., 2022)

[Weddell Sea polynya analysis using SMOS–SMAP apparent sea ice thickness retrieval](#)

*"Most of the content in this section should be in the discussion not in the conclusions"*

## Content Guidelines for Discussion Sections

*Discussion sections should begin with key findings, compare results with existing studies while explaining differences, and maintain logical flow throughout. Authors should emphasize negative findings, acknowledge limitations, and clearly state what questions remain unanswered. (1 source)*

Effective discussion sections follow a structured approach that begins with a brief summary of the most important findings before moving into detailed interpretation ([Jawaid et al., 2019](#)). The core of the discussion involves systematic comparison of results with local, regional, national, and international studies, which typically generates the majority of citations in this section ([Jawaid et al., 2019](#)). When findings align with previous research, authors should acknowledge these consistencies, but divergent results require more detailed attention and explanation ([Jawaid et al., 2019](#)). Potential reasons for differences might include variations in methodology, study populations, environmental conditions, or other contextual factors that should be explored thoroughly ([Jawaid et al., 2019](#)).

Authors should avoid repeating information already presented in the introduction and instead focus on what their study contributes to the scientific literature and how it advances understanding of the research topic ([Jawaid et al., 2019](#)). Negative findings deserve particular emphasis, as they often provide more valuable insights than expected positive results, and authors should resist the temptation to justify unexpected outcomes to match desired results ([Jawaid et al., 2019](#)). The discussion should maintain logical continuity and natural flow, guiding readers step-by-step through the reasoning without abrupt transitions ([Jawaid et al., 2019](#)). The most critical messages should appear at the beginning and end of the discussion, with the conclusion carrying a clear take-home message that honestly addresses whether the study objectives were met and what further research is needed ([Jawaid et al., 2019](#)). Study limitations should be included within this section, providing transparency about the research's scope and constraints ([Jawaid et al., 2019](#)).

## Evidence

(Jawaid et al., 2019)

[How to write introduction and discussion](#)

*"While writing discussion, start with a brief summary of your important findings. Then compare your results with local, regional, national and international studies. This is the most important part of a paper hence most of the references will come in this section while comparison is made with other studies. If your findings are similar which have been shown in earlier studies mention that but if they are not similar, it is more important. In that case one should try to find out the reason why your results are different. It is quite likely that there may be some difference in inclusion and exclusion criteria, there could be some cultural differences, differences in*

dietary pattern etc., Hence, try to find out those reasons and then elaborate that in detail. [1] Do not repeat what has already been stated in the introduction. Before writing discussion, sort out the important studies which you are going to cite in this section which have some important messages. Highlight what your study has added to the medical literature and how it will help in better understanding of the subject. Point out what further research is needed. Negative findings are more important than positive. If your study has failed to give expected answers, do not try to justify succumbing to the temptation of the desired results"

"Convey the message using logical arguments. Make sure that there is continuity and natural flow, no jumps, leading the reader step by step on the way"

"The most important message is always in the beginning of the discussion and at the end. Conclusion at the end of the discussion should carry the take home message. Be honest to tell the readers if the problem has been solved and the study answers the question posted in the objective of the study. Do mention if further studies are needed to confirm your observations, findings. Limitations of the study should be included in this section under separate sub-heading."

## Quality Standards and Evidence-Based Practices

Geoscience communication research increasingly emphasizes rigorous evaluation using social science methods and quantitative evidence rather than anecdotal reporting. Authors should write clearly for interdisciplinary audiences and systematically address limitations of new methods in discussion sections. (2 sources)

The field of geoscience communication has moved toward more rigorous, evidence-based practices that require systematic evaluation and assessment using established social science methods rather than relying solely on anecdotal observations (Hillier et al., 2021). This shift reflects a broader recognition that effective scientific communication, including the writing of discussion sections, benefits from empirical approaches that can be measured and validated. Research articles should incorporate qualitative and quantitative evidence to support their interpretations and conclusions (Hillier et al., 2021).

Given the interdisciplinary nature of geoscientific research, authors must write clearly and concisely while avoiding field-specific jargon that might exclude readers from other disciplines (Hillier et al., 2021). This accessibility requirement is particularly important in discussion sections, where complex interpretations must be communicated to diverse audiences spanning geology, chemistry, physics, environmental science, and related fields.

A specific evidence-based practice involves systematically addressing method limitations within discussion sections, particularly when introducing new analytical approaches or techniques (Xu et al., 2025). These limitations should be distilled from the discussion content and clearly articulate potential constraints of proposed solutions, providing readers with a realistic assessment of the method's scope and applicability (Xu et al., 2025). This practice enhances transparency and helps other researchers make informed decisions about adopting or adapting the methods for their own studies.

## Evidence

(Hillier et al., 2021)

[Editorial: Geoscience communication – Planning to make it publishable](#)

"This formalization included a call for increasingly robust evaluation and assessment of geoscience communication efforts through the use of evaluation instruments and social science methods"

"the current editors of GC, though, see the value of exploring the core aspects of what rigorous, evidenced-based geoscience communication research can look like"

"all research articles should include qualitative and/or quantitative evidence, and not solely anecdotal reporting"

"Write your paper (Section 10). Remember that the audience of GC spans many fields and disciplines. When writing your paper, please endeavour to write clearly and concisely, avoid jargon (see e.g. Venhuizen et al., 2019) and include critical structural elements you will be familiar with (e.g. introduction, methods, results, discussion, conclusions)."

(Xu et al., 2025)

[EarthSE: A Benchmark for Evaluating Earth Scientific Exploration Capability of LLMs](#)

*"New Method Limitations (L 1 ): Distilled from discussion sections, anticipating potential constraints of proposed solutions."*